



MISSOURI

S&T

MISSOURI UNIVERSITY OF SCIENCE AND TECHNOLOGY

Founded 1870 | Rolla, Missouri

Section 3.1

Introduction to Determinants

The Determinant of a 2×2 Matrix

Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. The determinant of A is

$$\det A = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

The Determinant of a 3×3 Matrix

Let $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$.

The determinant of A is

$$\det A = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$$

Example 1

Compute the determinant.

$$\begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 10 \end{vmatrix}$$

Submatrices and Cofactors

For any square matrix A , define A_{ij} as the submatrix formed by deleting row i and column j from A .

The (i, j) cofactor of A is $C_{ij} = (-1)^{i+j} \det A_{ij}$.

Sign Pattern:

+	-	+	...
-	+	-	
+	-	+	
⋮			⋱

The Determinant of an $n \times n$ Matrix

The determinant of an $n \times n$ matrix A can be computed by a cofactor expansion across any row or down any column of A .

The expansion across the i th row is

$$\det A = a_{i1}C_{i1} + a_{i2}C_{i2} + \cdots + a_{in}C_{in}$$

The expansion across the j th column is

$$\det A = a_{1j}C_{1j} + a_{2j}C_{2j} + \cdots + a_{nj}C_{nj}$$

Example 2

Compute the determinant.

$$\begin{vmatrix} 3 & 0 & 0 & 0 \\ 7 & -2 & 0 & 0 \\ 2 & 6 & 3 & 0 \\ 3 & -8 & 4 & -3 \end{vmatrix}$$

Determinants of Triangular Matrices

If A is a triangular matrix, then $\det A$ is the product of the entries on the main diagonal of A .

Example 3

Compute the determinant.

$$\begin{vmatrix} 3 & 0 & 0 & 0 \\ 3 & -8 & 4 & -3 \\ 2 & 6 & 3 & 0 \\ 7 & -2 & 0 & 0 \end{vmatrix}$$

Example 4

Compute the determinants.

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix}$$

$$\begin{vmatrix} ka & kb \\ c & d \end{vmatrix}$$

Numerical Note

In general, a cofactor expansion requires more than $n!$ multiplications.

The computation of a 25×25 determinant by cofactors would require approximately 1.5×10^{25} multiplications.

If a computer performs one trillion multiplications per second, this would take more than 500,000 years to compute.
